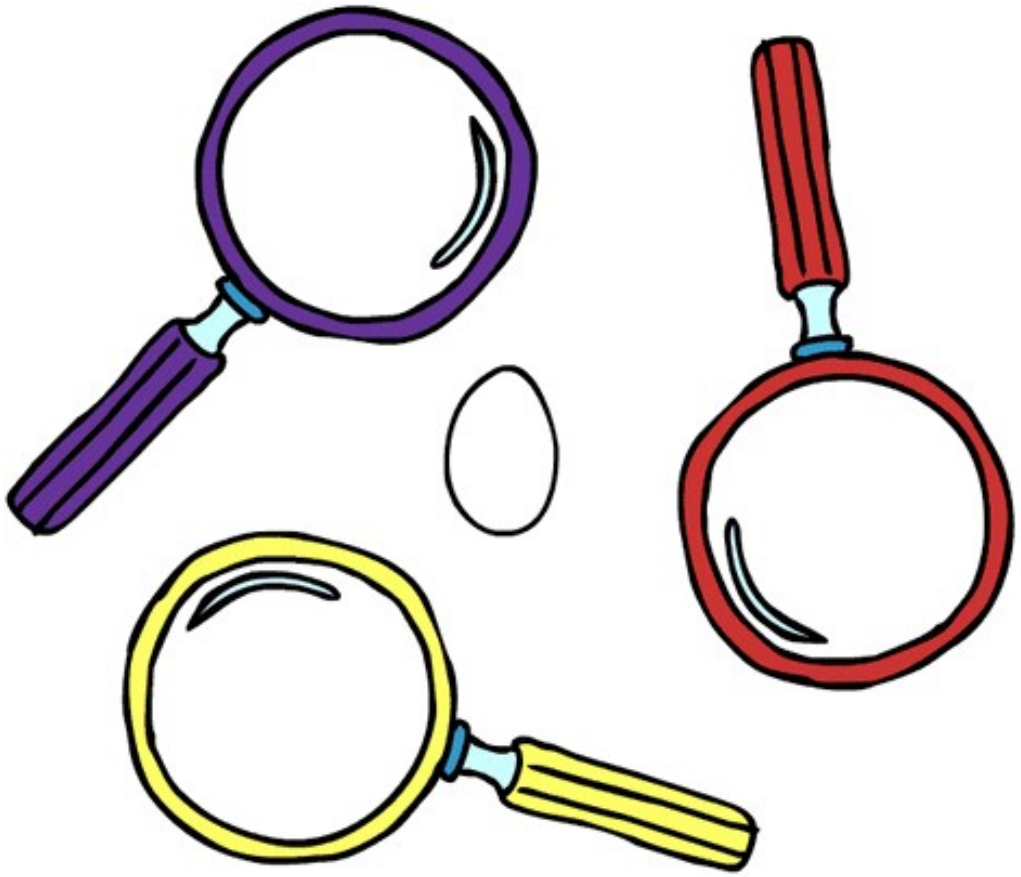


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Objectives



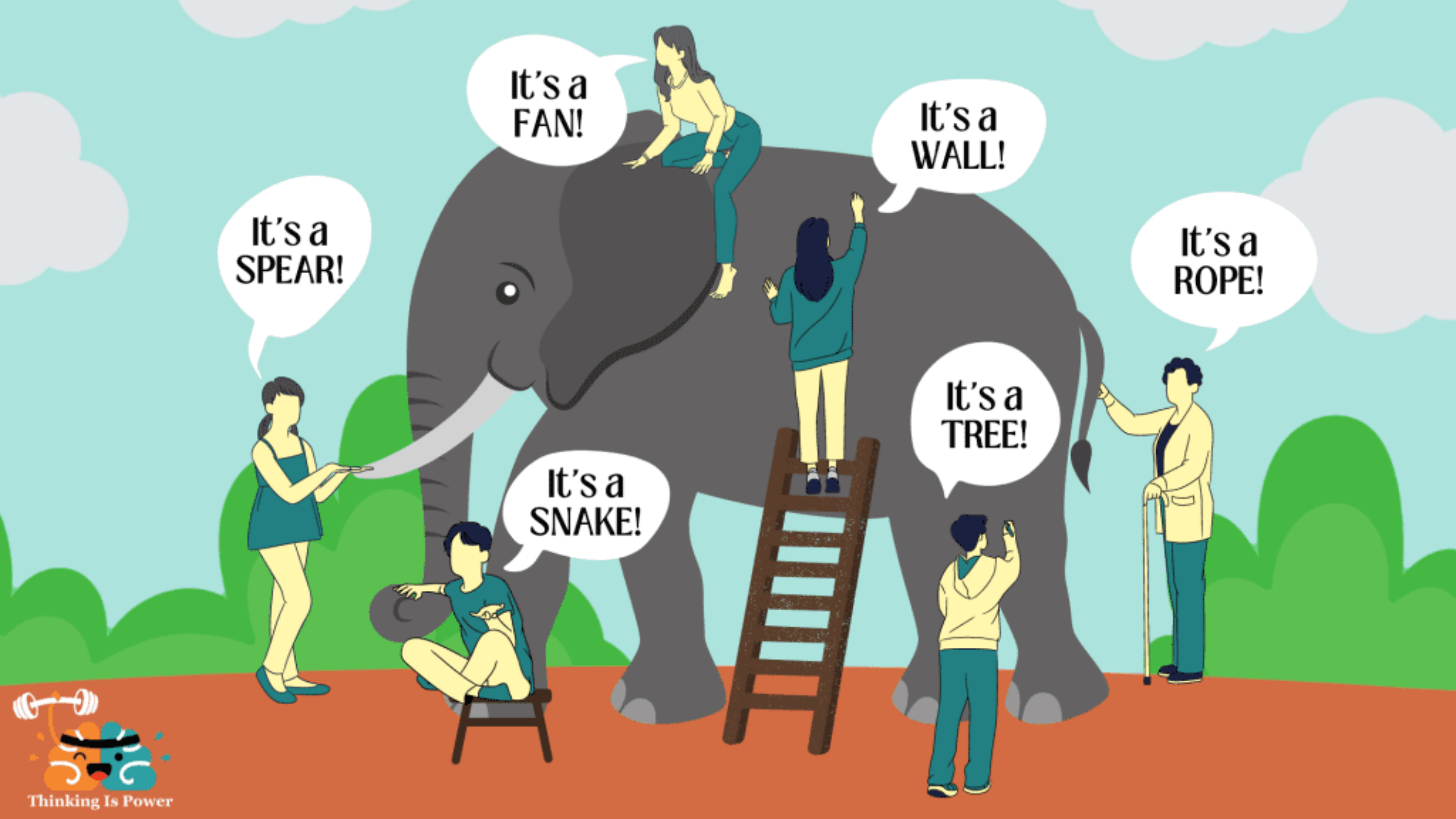
At the end of the lecture student should be able to:

- Define qualitative research and understand its purpose in health and dental sciences.
- Differentiate between qualitative and quantitative research approaches.
- Identify key characteristics of qualitative research, such as its focus on experiences, perceptions, and meanings.
- Describe common qualitative research methods, including interviews, focus groups, and observations.

Qualitative Research

- Qualitative research is a method of inquiry that focuses on understanding human experiences, behaviors, emotions, and perceptions through non-numerical data such as interviews, observations, and open-ended questions.

- It aims to explore the "why" and "how" behind certain phenomena rather than just measuring "how many" or "how much.


It's a FAN!
It's a WALL!
It's a SPEAR!
It's a ROPE!
It's a TREE!
It's a SNAKE!
Thinking Is Power

Qualitative Research: Why???



- **Understanding Patient Perspectives:** It helps uncover patients' beliefs, fears, motivations, and experiences related to oral health, treatment, and dental care

Example: How do patients describe their emotional experiences during a dental visit?

- **Exploring Health Behaviors:** Qualitative research provides insight into why peop adopt or avoid certain dental hygiene practices or treatments.

Example: What are the reasons why some individuals avoid regular toothbrushing despite being aware of its benefits?

- **Improving Communication:** It aids in developing better patient-dentist communication strategies by understanding how patients perceive health messages or dental visits.

Example: How do patients perceive the clarity and effectiveness of communication from their dentists during treatment planning?

Qualitative Research: Why???



- **Developing Patient-Centered Care:** Findings can guide the design of c
  that is more personalized, empathetic, and responsive to individual needs.

Example: What are patients' expectations and preferences for personalized dental care in a clinical setting?

- **Identifying Barriers to Care:** It helps explore social, cultural, and economic facto
  that prevent patients from accessing or adhering to dental care.

Example: What social and cultural factors do low-income patients perceive as barriers to accessing routine dental services?

- **Informing Health Policy and Education:** Qualitative insights are valuable in shaping dental education curricula, public health campaigns, and policy making by addressing real-world patient concerns and experiences.

Example: How do dental students perceive the effectiveness of their curriculum in preparing them for real-world patient interactions?


QUANTITATIVE RESEARCH
QUALITATIVE RESEARCH

Feature	Qualitative Research	Quantitative Research
Nature of Data		
Purpose		
Research Question		
Data Collection Methods		
Data Analysis		
Outcome		
Sample Size		
Role of Researcher		
Use in Dental Sciences		



Qualitative Research: Attributes



- **Exploratory and Descriptive:**

It seeks to explore "how" and "why" people think, feel, or behave in certain ways, rather than just measuring outcomes.

- **Non-Numerical Data:**

Uses words, narratives, images, and observations instead of numbers and statistics.

- **Open-Ended Approach:**

Data is collected through open-ended questions that allow participants to express themselves freely.

- **Small, Purposeful Samples:**

Focuses on specific individuals or groups who can provide rich and relevant information—not large random samples.

Qualitative Research: Attributes



- **Contextual Understanding:**

Emphasizes understanding behavior or experiences within the social, cultural, or personal context.

- **Subjective Interpretation:**

The researcher plays an active role in interpreting meanings and identifying themes in the data.

- **Flexible and Adaptive:**

The research process may evolve as new insights emerge during data collection and analysis.

Qualitative Research: Attributes



- **Thematic Analysis:**

Data is analyzed by identifying recurring themes, patterns, or concepts rather than using statistical tests.

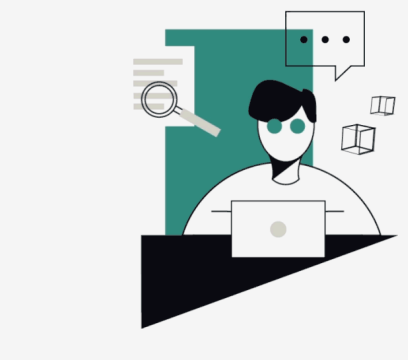
- **Participant-Centered:**

Gives voice to the perspectives, experiences, and emotions of participants.

- **Natural Settings:**

Often conducted in real-life environments (e.g., clinics, communities) rather than controlled laboratories.

Qualitative Research: Methods



- Interviews

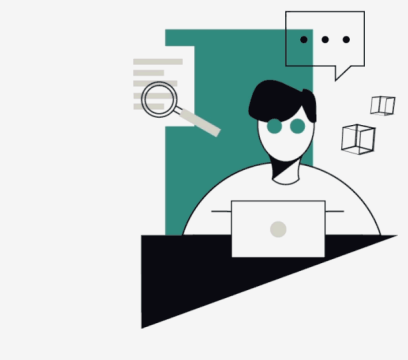
➢ One-on-one conversations to explore participants' views and experiences.

➢ Structured, semi-structured, or unstructured.

➢ Ideal for exploring individual perspectives in depth.

Advantages	Disadvantages
Allows deep exploration of personal experiences and insights	Time-consuming to conduct
Flexible—interviewer can follow up on interesting points	Risk of interviewer bias or leading questions
Good for sensitive topics in private settings	Relies heavily on participants' ability to articulate their thoughts

Qualitative Research: Methods



- Focus Groups

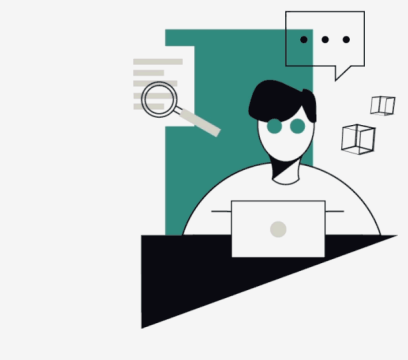
➢ Group discussions guided by a moderator to explore shared experiences and attitudes.

➢ Typically 6–10 people.

➢ Useful for understanding group norms, interactions, and collective opinions.

Advantages	Disadvantages
Encourages interaction and idea sharing among participants	Dominant voices may overshadow others
Can generate a wide range of views in a short time	Participants may be reluctant to share sensitive information in a group
Efficient for collecting data from multiple people simultaneously	Group dynamics may influence individual responses (e.g., peer pressure)

Qualitative Research: Methods



- Observations

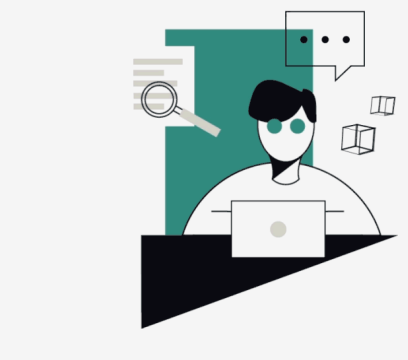
➢ Watching and recording behaviors and interactions in natural settings.

➢ Participant (researcher is involved) or non-participant.

➢ Effective for studying real-life settings and social practices.

Advantages	Disadvantages
Captures actual behavior in real-life settings	Observer presence may influence behavior (Hawthorne effect)
Provides context-rich and non-verbal data	Can be time-consuming and labor-intensive
Useful when participants cannot clearly express experiences verbally	Interpretation may be subjective or biased

Qualitative Research: Methods



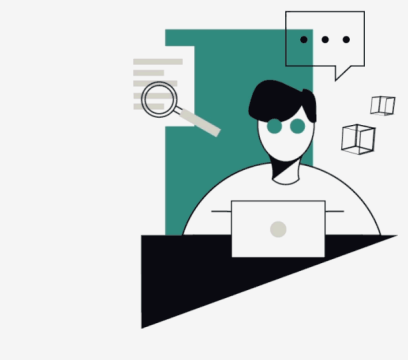
- Case Studies

➢ In-depth analysis of a single case (e.g., a person, group, event, or organization).

➢ Useful for gaining detailed contextual insights and understanding complex issues.

Advantages	Disadvantages
Offers deep, detailed understanding of complex phenomena	Limited generalizability to other contexts
Useful for exploring rare or unique situations	Can be time-consuming to collect and analyze large volumes of data
Allows multiple data sources (interviews, documents, observations)	Risk of over emphasis on one case

Qualitative Research: Methods



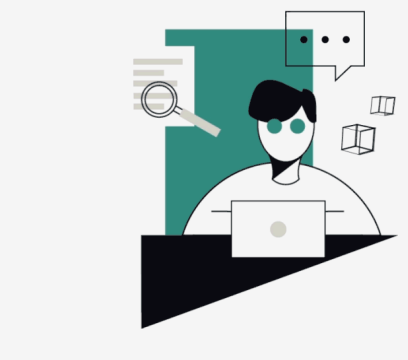
- Document or Content Analysis

➢ Systematic analysis of text-based or visual materials (e.g., reports, social media posts, diaries).

➢ Helps uncover underlying themes or patterns in communication

Advantages	Disadvantages
Useful for historical or retrospective analysis	Limited to available or accessible documents
Can be unobtrusive—does not require direct interaction with participants	May lack context or explanation behind content
Enables analysis of trends over time	Interpretation may be subjective without triangulation

Qualitative Research: Methods



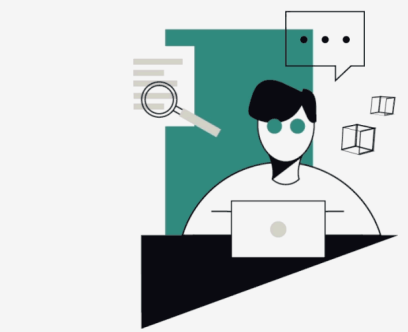
- Ethnography

➢ Extended study of people and cultures in their natural environment, often involving immersion.

➢ Common in anthropology and sociology to understand cultural practices and beliefs.

Advantages	Disadvantages
Provides deep cultural insights through long-term immersion	Requires significant time and researcher involvement
Captures practices and meanings in real-life contexts	Researcher may "go native" and lose objectivity
Offers holistic understanding of a group or community	Ethical issues related to informed consent and privacy

Qualitative Research: Methods



- Narrative Research

➢ Examines personal stories to understand how individuals make sense of their experiences.

➢ Helps reveal how people construct meaning through storytelling.

Advantages	Disadvantages
Reveals how individuals make meaning of their experiences through storytelling	Time-intensive to collect and analyze narratives
Emphasizes individual voices and perspectives	Findings may be difficult to generalize
Useful in understanding identity, life transitions, and cultural context	May be influenced by memory recall or storytelling style



- **What is the primary focus of qualitative research?**

A. Measuring numerical outcomes
B. Understanding human experiences and perceptions
C. Testing hypotheses using statistics
D. Conducting randomized trials

✅ Answer: B

- **Which of the following is NOT a common method of qualitative research?**

A. Case study
B. Focus group
C. Laboratory experiment
D. Ethnography

✅ Answer: C

- **Qualitative research is particularly useful for studying:**

A. Drug efficacy
B. Behavior in natural settings
C. Genetic mutations
D. Disease prevalence

✅ Answer: B

- **In qualitative research, the role of the researcher is:**

A. Passive observer
B. Completely objective
C. Active interpreter
D. Data processor

✅ Answer: C

- **Which qualitative method involves analyzing personal stories?**

A. Ethnography
B. Narrative research
C. Focus group
D. Document analysis

✅ Answer: B

- **A focus group typically includes how many participants?**

A. 2–4
B. 6–10
C. 15–20
D. Over 30

✅ Answer: B

- **What type of sampling is common in qualitative research?**

A. Random sampling
B. Stratified sampling
C. Purposive sampling
D. Systematic sampling

✅ Answer: C

- **Thematic analysis in qualitative research involves:**

A. Statistical hypothesis testing
B. Coding and identifying patterns in data
C. Comparing means and medians
D. Randomizing participants into groups

✅ Answer: B

- **Which method is best for studying interactions in a natural setting?**

A. Interviews
B. Case studies
C. Observations
D. Surveys

✅ Answer: C

- **Which is a key characteristic of qualitative research?**

A. Use of control groups
B. Emphasis on objectivity
C. Use of non-numerical data
D. Focus on generalizability

✅ Answer: C

- **Which qualitative method is most suitable for studying cultural beliefs and practices?**

A. Ethnography
B. Narrative research
C. Content analysis
D. Case study

✅ Answer: A

- **Document analysis is primarily concerned with:**

A. Lab results
B. Patient interviews
C. Visual or text-based materials
D. Questionnaire data

✅ Answer: C

- **One strength of focus groups is:**

A. Individual depth
B. Group dynamic insights
C. Complete anonymity
D. High generalizability

✅ Answer: B

- **A major weakness of observations is:**

A. Limited real-life context
B. Risk of Hawthorne effect
C. Poor depth of data
D. Lack of flexibility

✅ Answer: B

- **Which of the following is TRUE about case studies?**

A. They involve large random samples
B. They provide shallow, quick insights
C. They allow detailed, in-depth understanding
D. They are usually experimental

✅ Answer: C

- **In qualitative research, open-ended questions are used to:**

A. Test hypotheses
B. Control variables
C. Measure frequency of events
D. Allow participants to freely express experiences

✅ Answer: D

- **One of the main purposes of qualitative research in dental education is to:**

A. Improve communication and patient-centered care
B. Compare drug efficacy
C. Understand tooth structure
D. Perform statistical validation

✅ Answer: A

- **Which of the following methods is ideal for exploring an individual's personal experience in depth?**

A. Focus group
B. Interview
C. Observation
D. Survey

✅ Answer: B

- **What is a common challenge in conducting focus groups?**

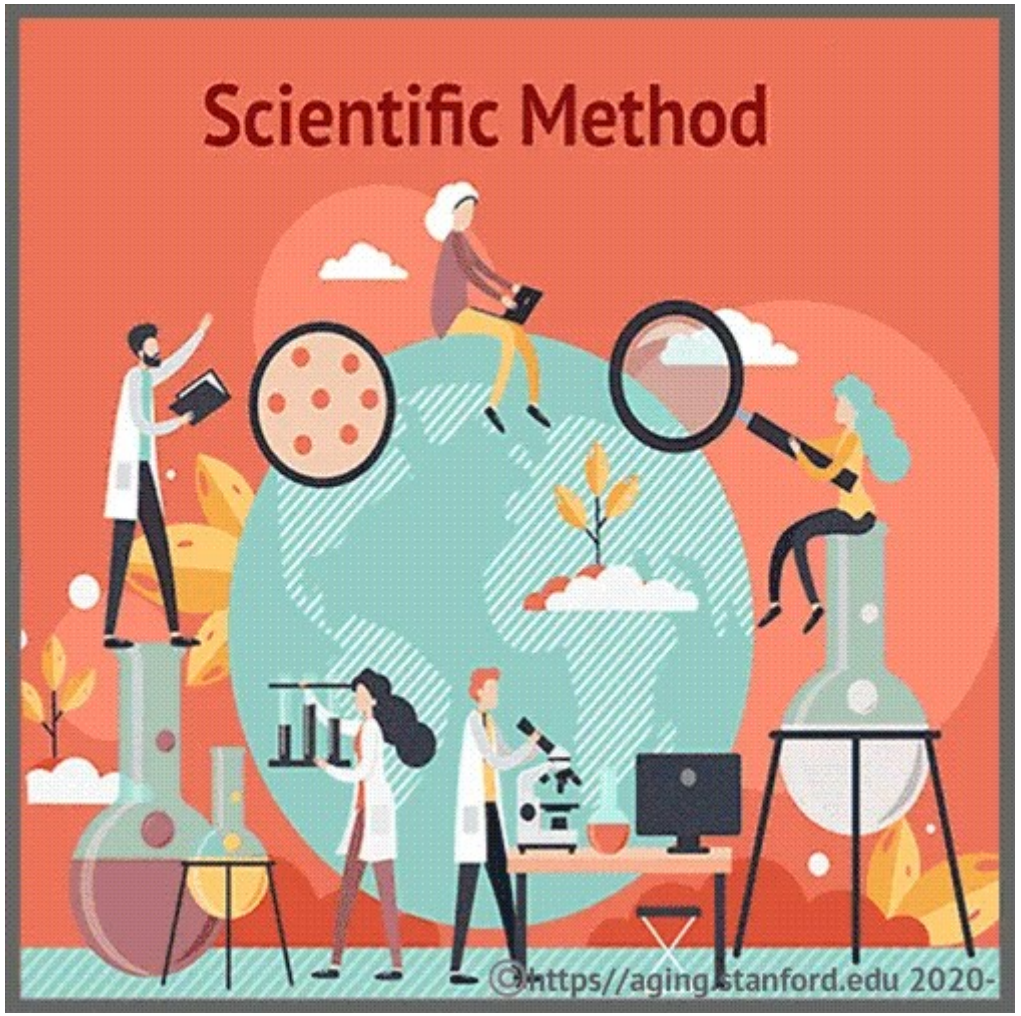
A. Lack of data
B. Inflexibility
C. Dominance of certain participants
D. No opportunity for discussion

✅ Answer: C

- **What type of data is typically generated in qualitative research?**

A. Percentages
B. Scores
C. Narratives and themes
D. Charts and graphs

✅ Answer: C


Scientific Method
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A researcher spends six months observing tribal dental hygiene practices while living in a rural village. Research Type?
Ethnography

A dental student interviews patients one-on-one to explore their fear of root canal treatments.
Interviews

Researchers hold a group discussion with 8 patients to explore their experiences with dental implants.
Focus Group

A study examines diaries written by patients documenting their post-operative dental pain for two weeks.
Document or Content Analysis

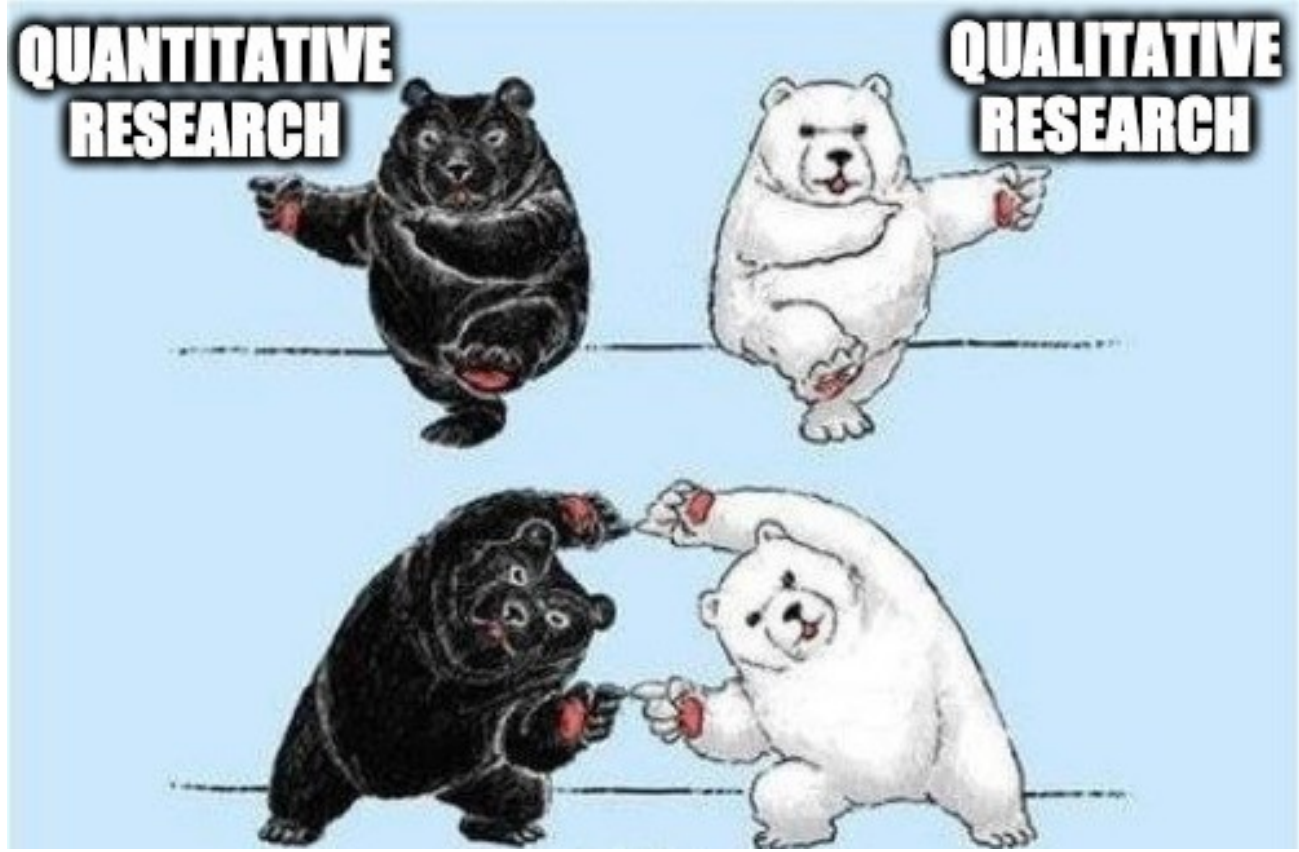
A researcher closely observes patient behavior in a dental waiting area without interacting.
Non-Participant Observation

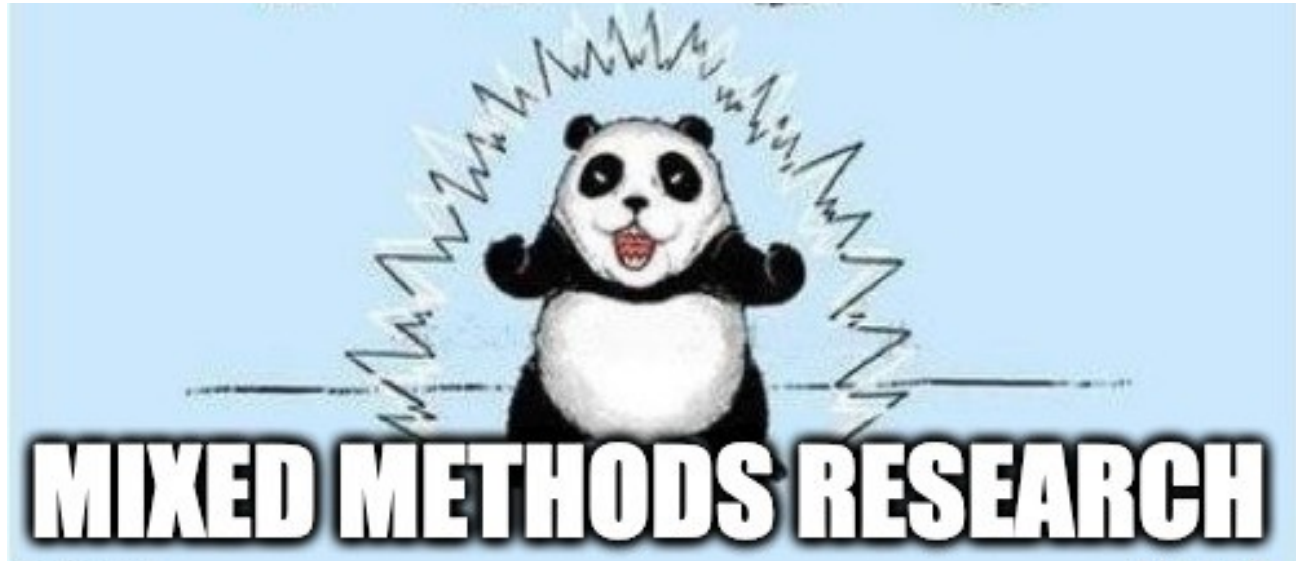
A professor conducts an in-depth study of one dental student’s experience during their clinical rotations
StudyCase 7:Patients narrate their journey of recovering from maxillofacial surgery, focusing on meaning-making.✅ Type: Narrative Research

Patients narrate their journey of recovering from maxillofacial surgery
Narrative Research

What are the cultural practices influencing patient care in a rural hospital?
Ethnography

How is a new teaching strategy implemented in a medical school?
Case study


QUANTITATIVE RESEARCH
QUALITATIVE RESEARCH


MIXED METHODS RESEARCH


THANK YOU